Hongyu Liu

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Education

The Chinese University of Hong Kong, Shenzhen 09/2026 –06/2028 (Expected)
Master's in Data Science

South China Normal University 09/2022 –06/2026
Bachelor's in Software Engineering GPA: 4.06/5 (Top 10%)

Technology Trends and Industry Analysis

Reasoning Compute in the Agent Era Blog
Maps GPU/ASIC, HBM, packaging, data-center, and cloud profit pools by effective task cost; tracks CapEx, contracts, financing, rents, utilization, renewals, and returns to spot overheating and profit shifts.

AI Agents and Enterprise Coordination Blog
Compares enterprise agent approaches from Grok Bot, Claude Tag, and Doubao Work Partner across context, permissions, handoffs, and acceptance; examines value in collaboration entry points and cross-system governance, with indicators for enterprise willingness to pay.

Safety Gaps in Speech LLMs Blog
Extends VoxPrivacy and VoxSafeBench research into Speech AI industry analysis, covering model capability, safety evaluation, and vertical applications; identifies potential opportunities in continuous evaluation infrastructure and high-value use cases.

Research and Projects

Trustworthy Speech LLM Evaluation 07/2025 –01/2026
CUHK-Shenzhen Speech Team Collaboration · Second Author · Benchmark Design & Build · Data Processing VoxPrivacy · ICLR
VoxSafeBench

- Built two benchmarks spanning multi-user interaction privacy, safety, and fairness. Cross-model evaluation found that frontier models detect speaker, tone, and environmental cues but struggle to use them reliably in privacy and safety decisions, exposing a gap between hearing cues and making sound decisions.

Audio Dialogue and Intent Recognition 02/2025 –06/2025
First Author · Data Processing · Model Design · Experimental Validation DialogGraph-LLM · ECAI Oral
MSMT-FN · ADMA

- Studied intent recognition in long, multi-speaker dialogue through multi-segment, multi-task fusion, relational dialogue modeling, and semi-supervised learning; evaluated how different approaches map raw audio to intent predictions.

TradingAgent | Multi-Agent Investment Research and Thesis Validation Framework 01/2026 –Present · GitHub
Restructured multi-agent analysis for A-share company research into a unified evidence chain across filings, financials, news, markets, and valuation; applied evidence checks, bull/bear cross-validation, and historical and peer valuation constraints to produce traceable theses, risks, and validation criteria

Internship Experience

Z Explorer Oct 26, 2026 –Present
Z Potentials · Intern | Remote

- **Early-Stage Technology Research:** Participate in industry and company research on early-stage technology, with a scope spanning in-depth article synthesis, interviews, and performing industry and product analysis.

Insta360 02/2026 –06/2026
Speech Algorithm Intern, Research Institute · Shenzhen

- **On-Device Speech Products and Model Evaluation:** Compared cascaded and end-to-end speech models for drone and assistant intent tasks, keeping models under 50M parameters at baseline; evaluated multimodal LLMs on image understanding, answer quality, simultaneous interpretation, and tool use.

Amphion Technology Co., Ltd. 06/2025 –09/2025
R&D Department · Shenzhen

- **Speech Product Development:** Developed core algorithms for Makaw, a voice-cloning app, including model training and business iterations; contributed ASR/TTS to a video-translation backend.

Skills and Honors

Honors: Provincial Second Prize, China Undergraduate Mathematical Contest in Modeling (2024) | Honorable Mention, MCM (2025) | National Second Prize, MathorCup (2024)
Skills: Python for data analysis | Agent tools including Codex and Claude Code | English (IELTS 6.5)
Activities: Staff Member, Youth League Committee Research and Survey Department | College Basketball Team Captain